# ANUP BAJGAIN

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Motivated aerospace engineering student passionate about solving complex problems. Quick to adapt, eager to grow, and committed to continuous improvement.

## EDUCATION

**Bachelor of Engineering in Aerospace Engineering** 2022–2026
IOE, Pulchowk Campus, Tribhuvan University, Lalitpur cumulate: 83.45%

***Relevant Coursework:*** Aerodynamics, Flight Dynamics, Propulsion Systems, Avionics, Control Systems, CFD, Aircraft Manufacturing Process, Aircraft Maintenance Process

## EXPERIENCE

**Operations Engineering Intern** *2025–2026 (6 months)*
*Yeti Airlines, Kathmandu, Nepal*

- Supported OCC and Dispatch operations, assisting with real-time flight monitoring and operational decision-making for the ATR 72-500 fleet.
- Assisted with Weight & Balance calculations, pre-flight documentation, and takeoff/landing performance analysis under varying operational conditions.
- Maintained operational records and technical documentation in compliance with CAAN regulatory requirements.

## PROJECTS

**Study of the Surface Heating Effect on a Compressible Wall Jet using Rainbow Schlieren Deflectometry**

- Conducted experimental flow visualization using Rainbow Schlieren Deflectometry (RSD) to measure density gradient fields in a compressible wall-jet configuration.
- Performed 3D RANS CFD simulations in ANSYS Fluent and validated results against experimental data.
- Developed Python post-processing scripts for quantitative comparison of CFD and experimental density gradient profiles across multiple streamwise stations.

**Agricultural UAV Design**

- Modeled the aircraft geometry in Plane Maker and performed aerodynamic validation using X-Plane flight simulator.
- Extracted and analyzed simulated flight data to verify aerodynamic performance metrics against analytical estimates.

## TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Matplotlib, OOP, DSA)
- **Analysis & Simulation:** ANSYS Fluent, OpenFOAM, XFLR5
- **CAD:** CATIA V5, SolidWorks
- **Others:** LaTeX, MS office

## ADDITIONAL

Certifications: 3D Design, Python Programming, Machine Learning
Interests: Music, Travel, Chess, Football